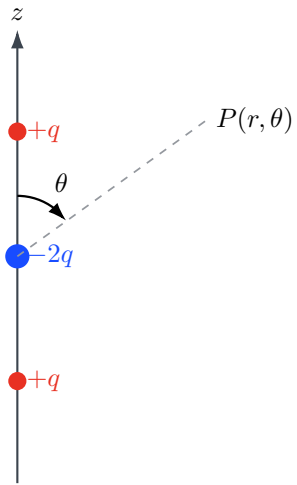
线型电四极子：零总电荷、零偶极矩，首项为四极项



$$\varphi = \frac{kq}{|\mathbf{r} - \ell \mathbf{e}_z|} - \frac{2kq}{r} + \frac{kq}{|\mathbf{r} + \ell \mathbf{e}_z|}$$

$$\varphi_2 = \frac{2kq\ell^2}{r^3} P_2(\cos\theta), \quad \frac{\varphi_4}{\varphi_2} = O(\ell^2/r^2)$$

$P_2(\cos\theta) = 0 \Rightarrow \cos^2\theta = 1/3$；
零势锥上通常仍有切向电场。